Integration — Lesson 15 Test: Two Special Functions, e^x and $1/x$

10 questions · show your work in the box · calculators allowed · give exact answers first (with e or ln), then decimals where asked · useful values: $e \approx 2.72$, $e^2 \approx 7.39$, $\ln 4 \approx 1.39$

Name: ______________________________ Date: ______________ Score: ______ / 10

1. Evaluate without a calculator: (a) $\ln 1$ (b) $\ln e$ (c) $\ln(e^4)$. For each, say in words what question ln is answering. 1 pt

2. Find $\int e^x \, dx$, and explain in one sentence why this is "the easiest antiderivative in the world." 1 pt

3. Find $\int (6/x) \, dx$ (for positive x). 1 pt

4. Find $\int (x^3 + 2/x)\,dx$. 1 pt

5. Evaluate $\int_1^{e^3} (1/x)\,dx$. 1 pt

6. Evaluate $\int_0^1 4e^x\,dx$. Exact answer first, then a decimal rounded to two places. 1 pt

7. The power rule says $\int x^n\,dx = x^{n+1}/(n+1) + C$. Explain why it cannot handle $1/x$, and state what $\int(1/x)\,dx$ actually equals. 1 pt

8. What is the special property of e^x that no other basic function has, and what does that property mean about a population whose size follows e^t? 1 pt

9. Find the function F with $F'(x) = 1/x$ and $F(1) = 7$. 1 pt

10. Evaluate $\int_1^e (3/x + 2x)\,dx$. Simplify to an exact answer, then give a decimal rounded to two places. 1 pt